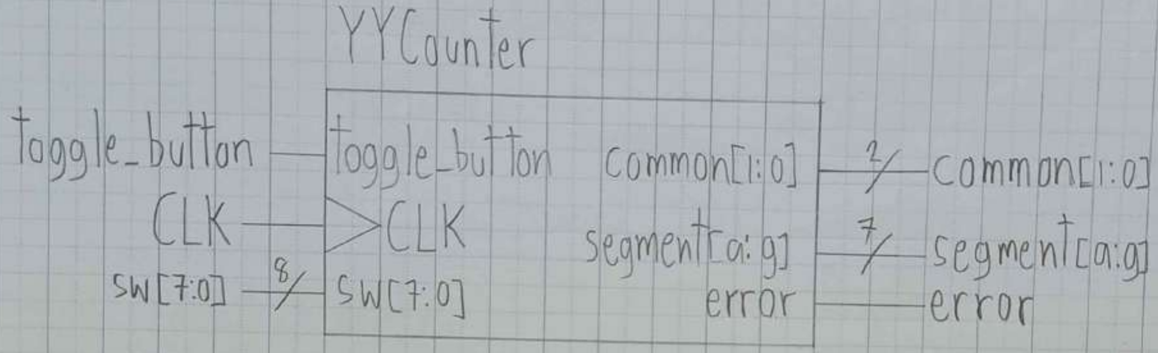
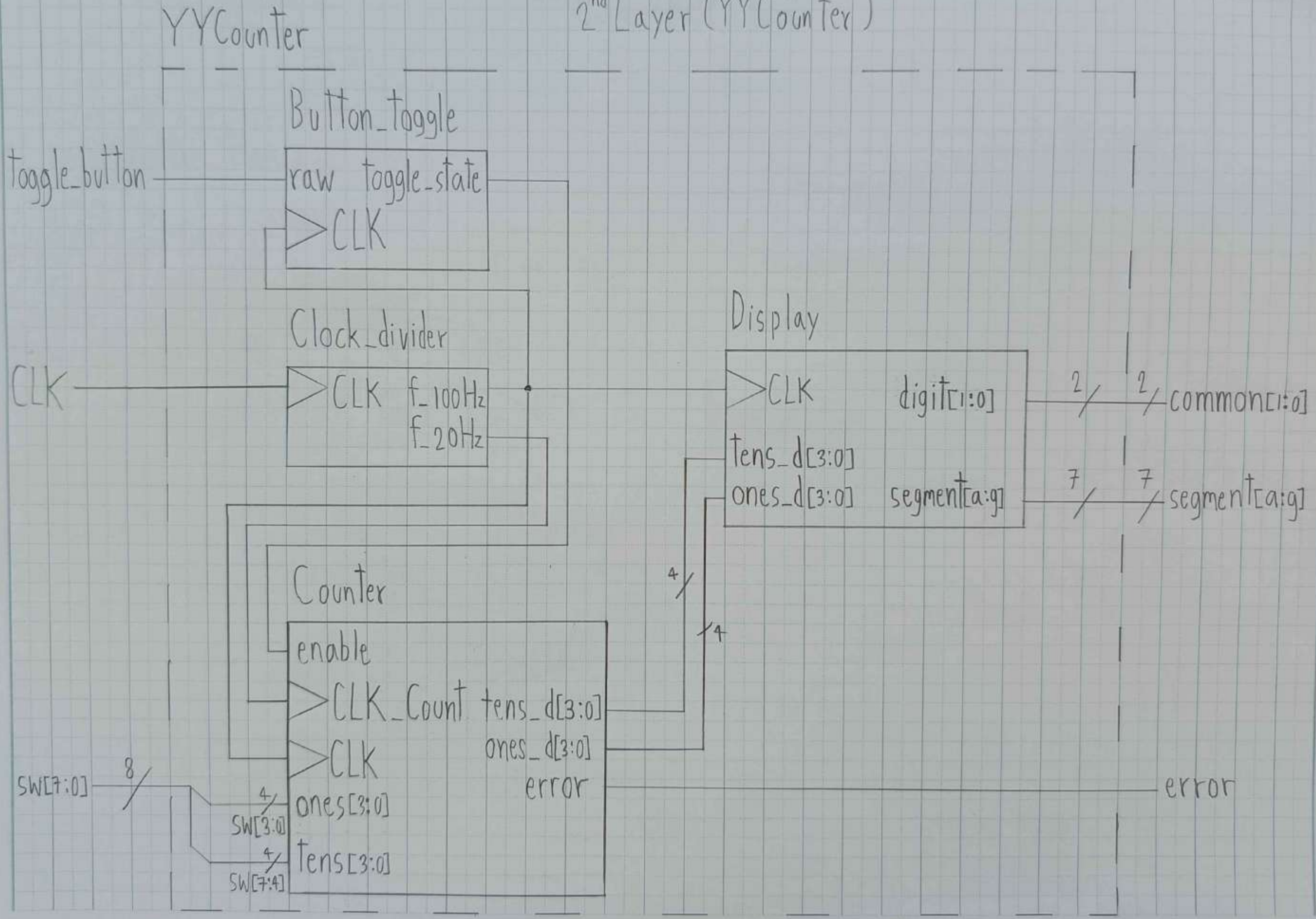
Top Layer (Lab6)


YYCounter
toggle_button
CLK
SW[7:0]
8
toggle_button
CLK
SW[7:0]
common[1:0]
segment[a:g]
error
2
7
common[1:0]
segment[a:g]
error

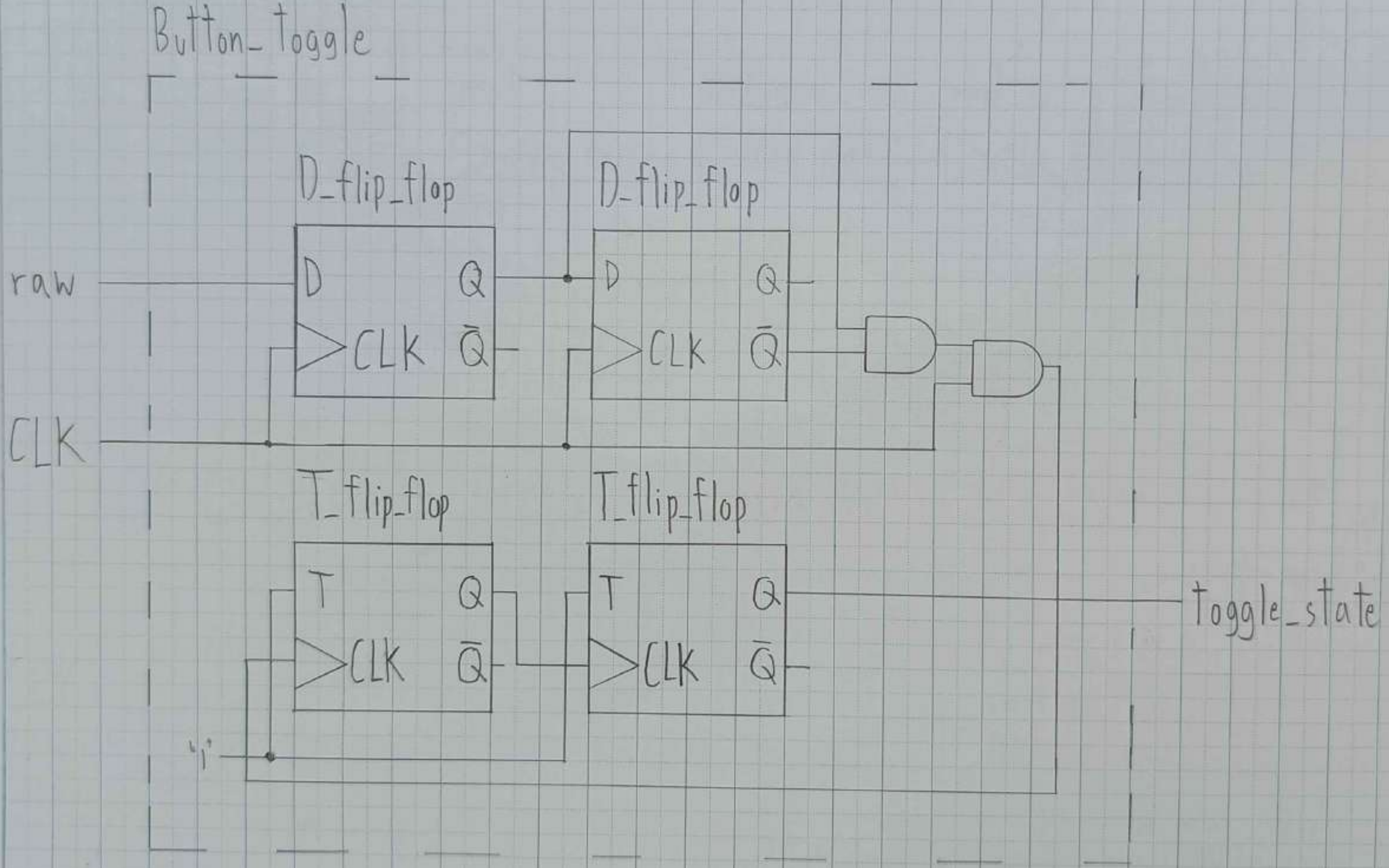
▽Lab6 นายอรุณ รักษ์วงศ์สิริ 68011008 sec 102

2nd Layer (YYCounter)


YYCounter
Button_toggle
toggle_button
raw
toggle_state
CLK
Clock_divider
CLK
CLK
f_100Hz
f_20Hz
Display
CLK
digit[1:0]
2
2
common[1:0]
tens_d[3:0]
ones_d[3:0]
segment[a:g]
7
7
segment[a:g]
4
4
Counter
enable
CLK_Count
CLK
tens_d[3:0]
ones_d[3:0]
error
error
SW[7:0]
8
4
ones[3:0]
SW[3:0]
4
tens[3:0]
SW[7:4]

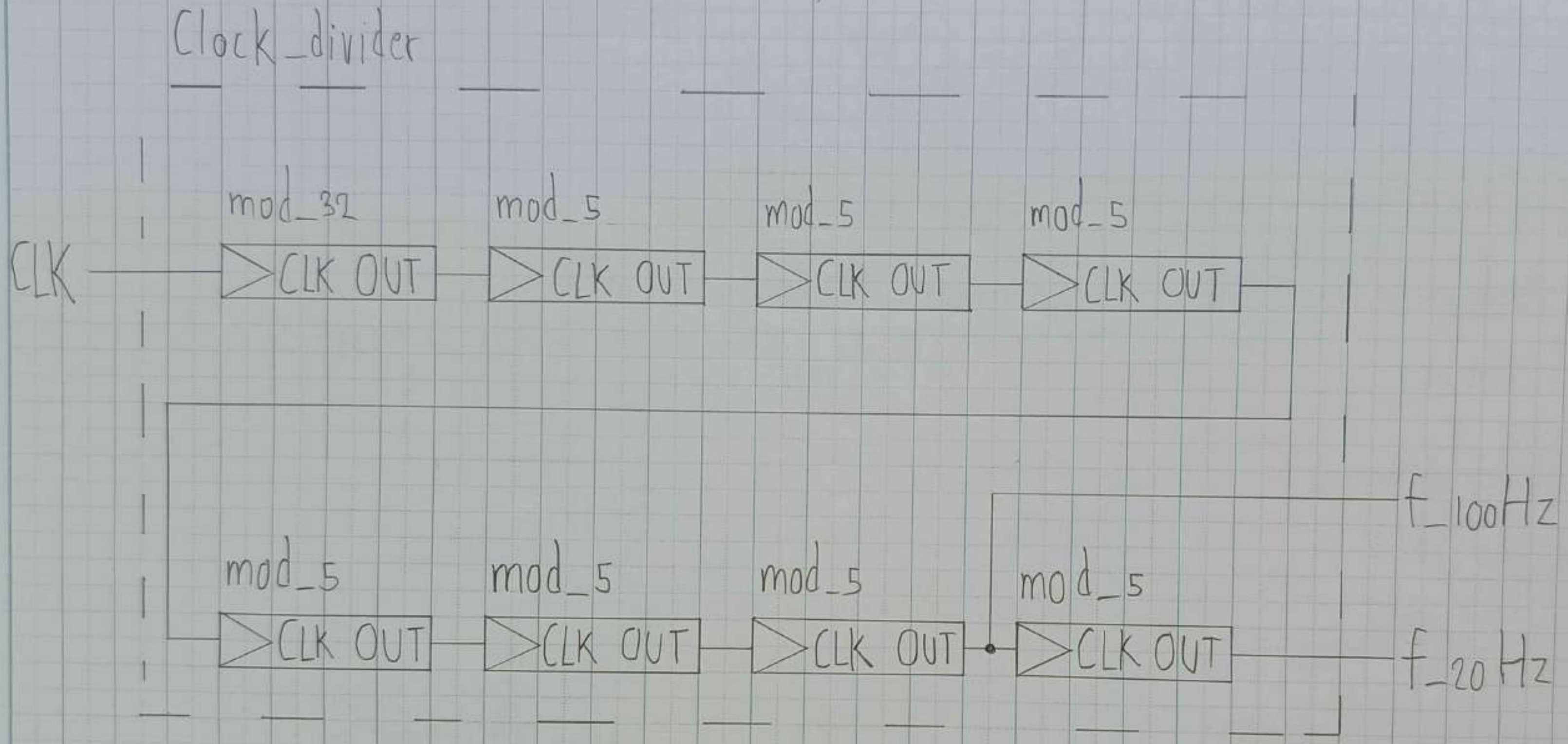
▽Lab6 นายอรุณ รักษ์วงศ์สิริ 68011008 sec 102

3rd Layer (Button_toggle)


Button_toggle
D_flip_flop
D_flip_flop
raw
D
Q
CLK
Q̄
D
Q
CLK
Q̄
CLK
T_flip_flop
T_flip_flop
T
Q
CLK
Q̄
T
Q
CLK
Q̄
toggle_state
"1"

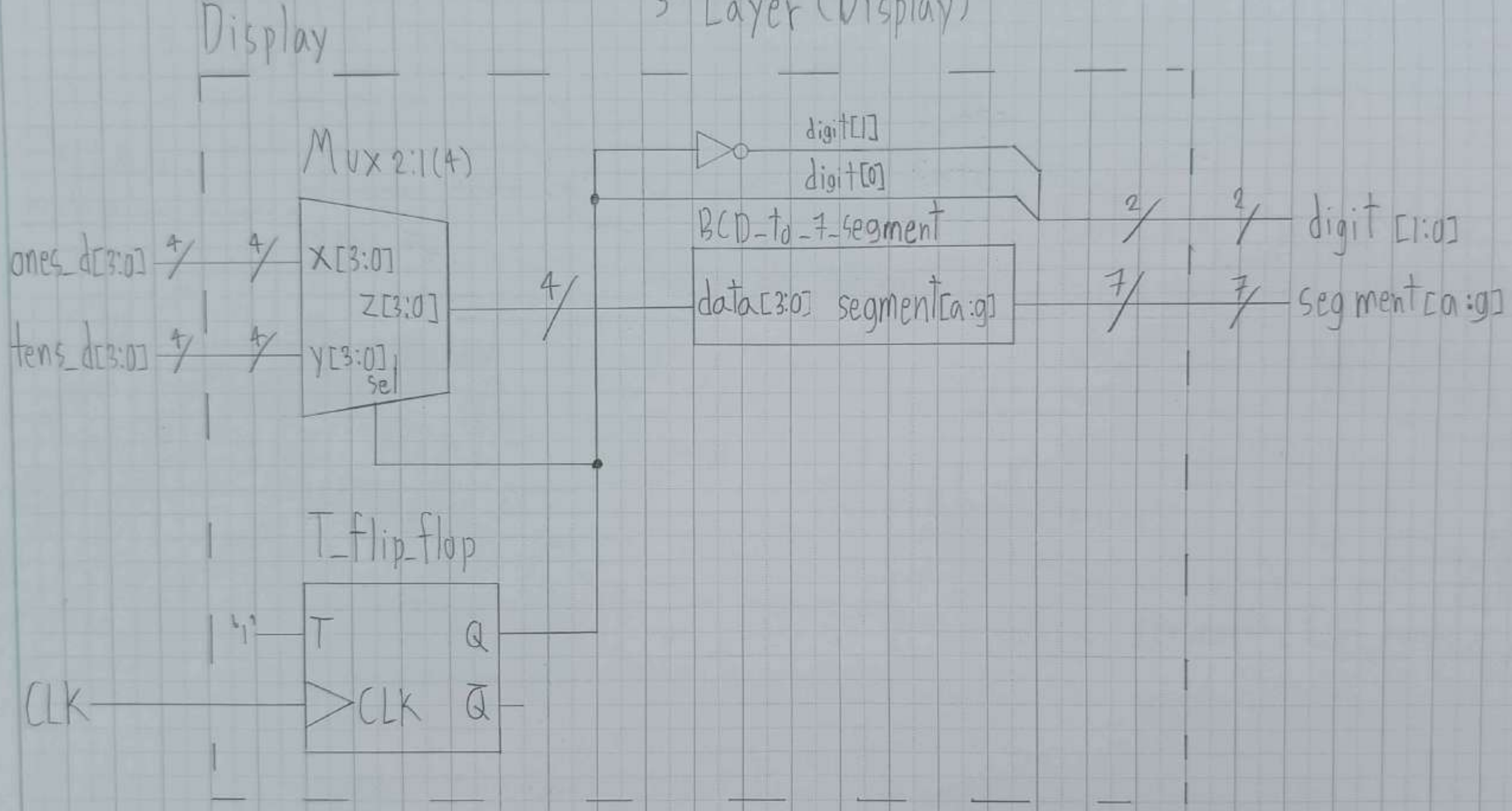
▽Lab 6 นายวรุฒ รักษ์วงศ์ สิริ 68011008 sec 102

3^{rd} Layer (Clock_divider)


Clock_divider
CLK
mod_32
mod_5
mod_5
mod_5
CLK OUT
CLK OUT
CLK OUT
CLK OUT
mod_5
mod_5
mod_5
mod_5
CLK OUT
CLK OUT
CLK OUT
CLK OUT
f_100Hz
f_20Hz

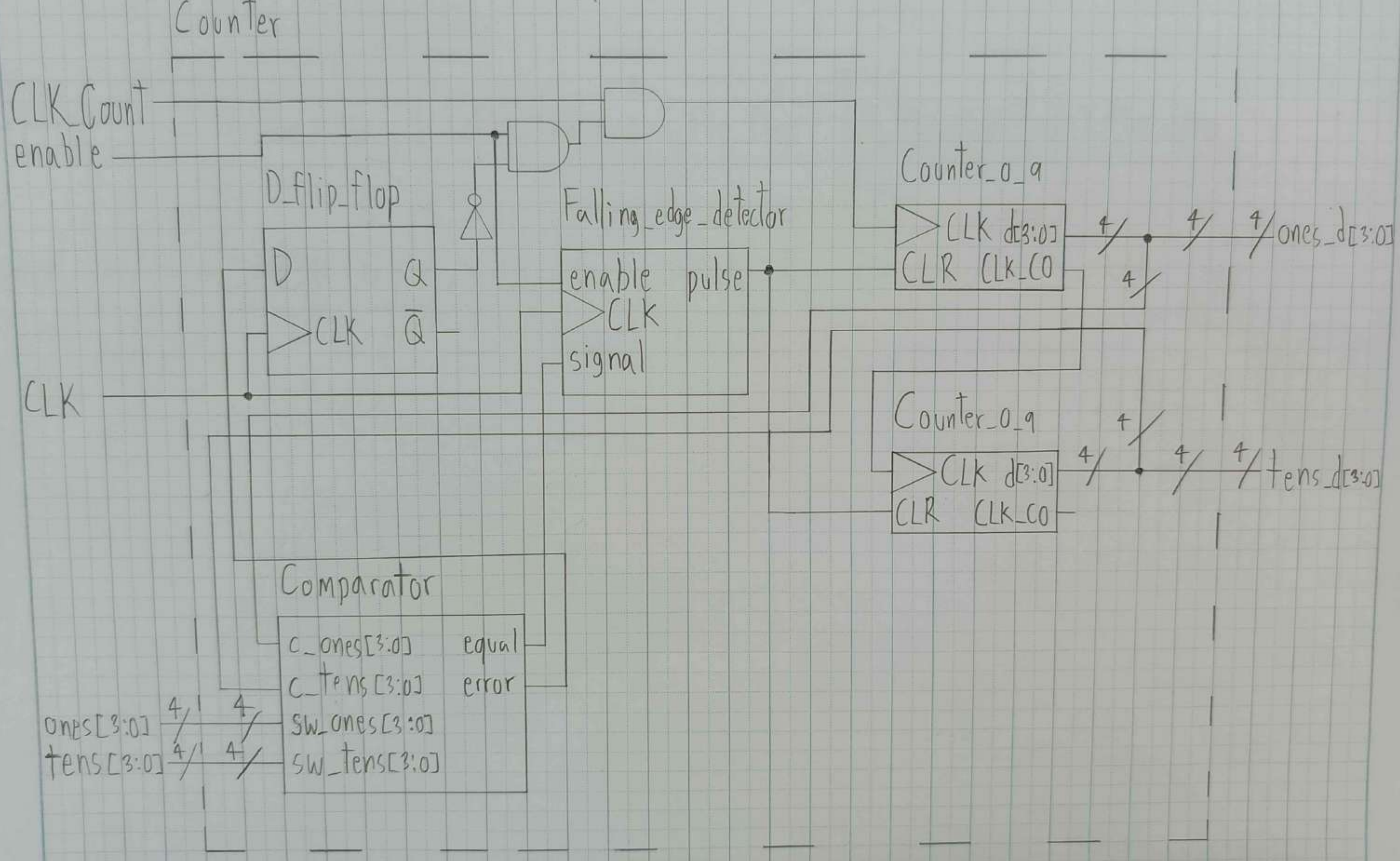
▽ Lab6 นายอรุณ รักษ์วงศ์สิริ 68011008 sec 102

3rd Layer (Display)


Display
Mux 2:1(4)
ones_d[3:0]
4
4
X[3:0]
Z[3:0]
4
tens_d[3:0]
4
4
Y[3:0]
sel
digit[1]
digit[0]
BCD_to_7_segment
data[3:0]
segment[a:g]
2
2
digit [1:0]
7
7
segment[a:g]
T_flip_flop
"1"
T
Q
CLK
CLK
Q̄

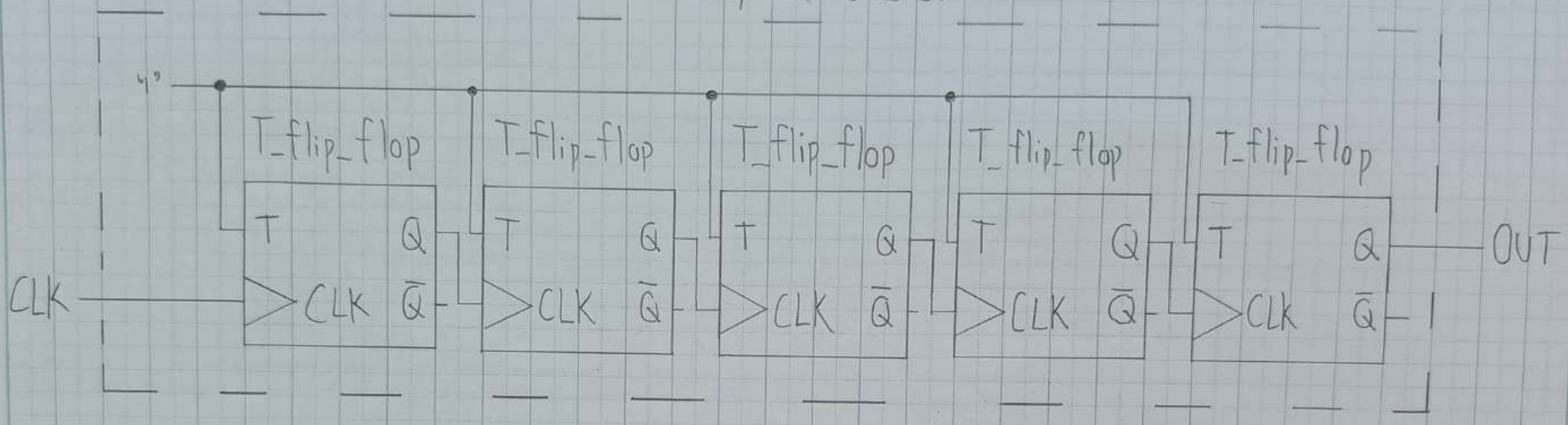
▽Lab6 นายวรุณ รักษ์ของศักดิ์ 68011008 sec 102

3rd Layer (Counter)


Counter
CLK_Count
enable
D_flip_flop
D
Q
CLK
Q̄
Falling_edge_detector
enable
pulse
CLK
signal
CLK
Counter_0_9
CLK
d[3:0]
CLR
CLK_CO
4
4
4
4
ones_d[3:0]
Counter_0_9
CLK
d[3:0]
CLR
CLK_CO
4
4
4
4
tens_d[3:0]
Comparator
c_ones[3:0]
equal
c_tens[3:0]
error
sw_ones[3:0]
sw_tens[3:0]
ones[3:0]
tens[3:0]
4
4
4
4

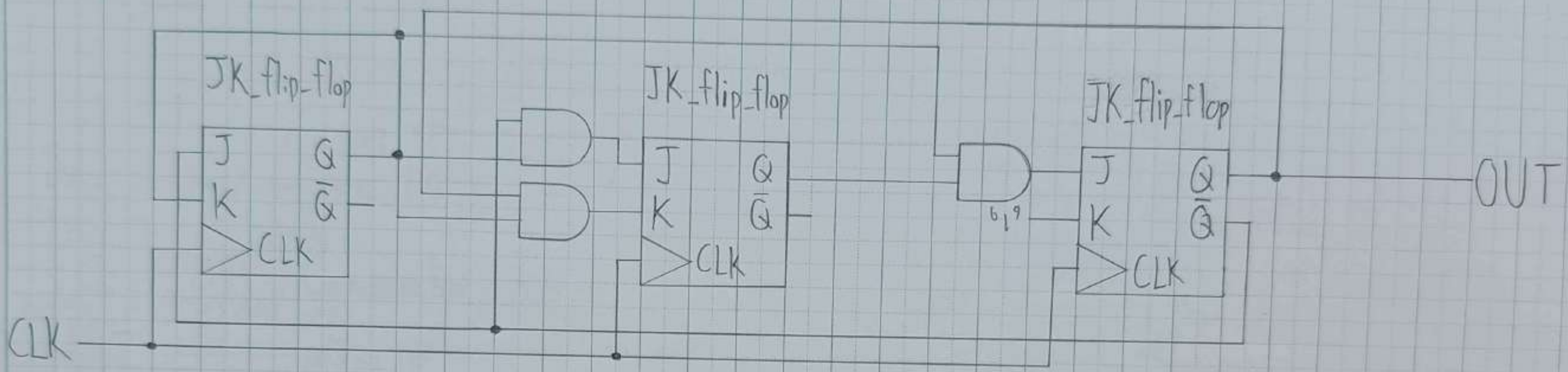
▽Lab6 นายอรุณ รักษ์วงศ์ษิริ 68011008 sec 102

4th Layer (mod_32)


"1"
T_flip_flop
T_flip_flop
T_flip_flop
T_flip_flop
T_flip_flop
T
Q
CLK
$\bar{Q}$
CLK
OUT

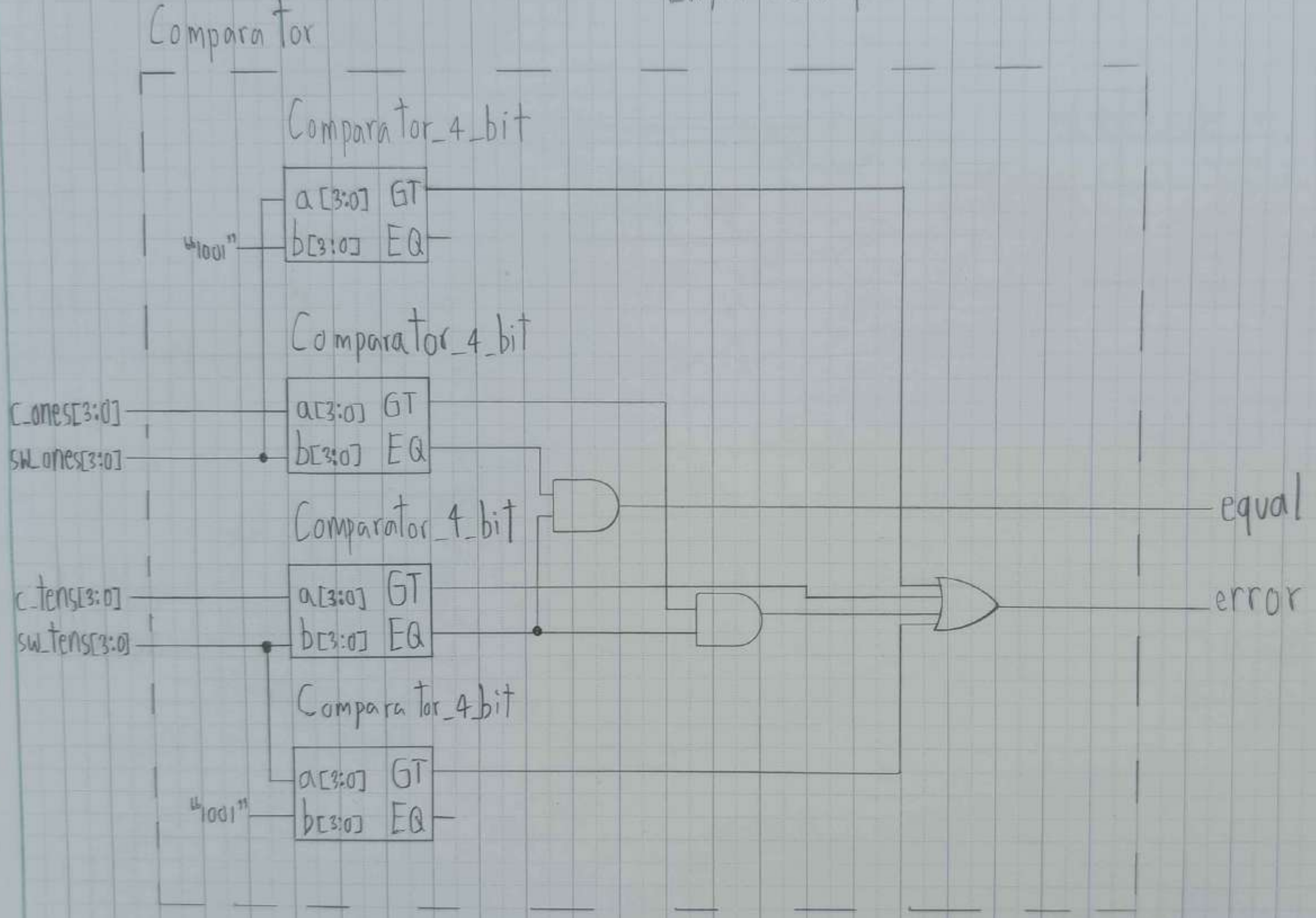
▽Lab6 นายอรุณ รักษ์วงศ์สิริ 68011008 sec102

4th Layer(mod_5)


JK_flip_flop
J
Q
K
Q̄
CLK
JK_flip_flop
J
Q
K
Q̄
CLK
6,9
JK_flip_flop
J
Q
K
Q̄
CLK
OUT
CLK

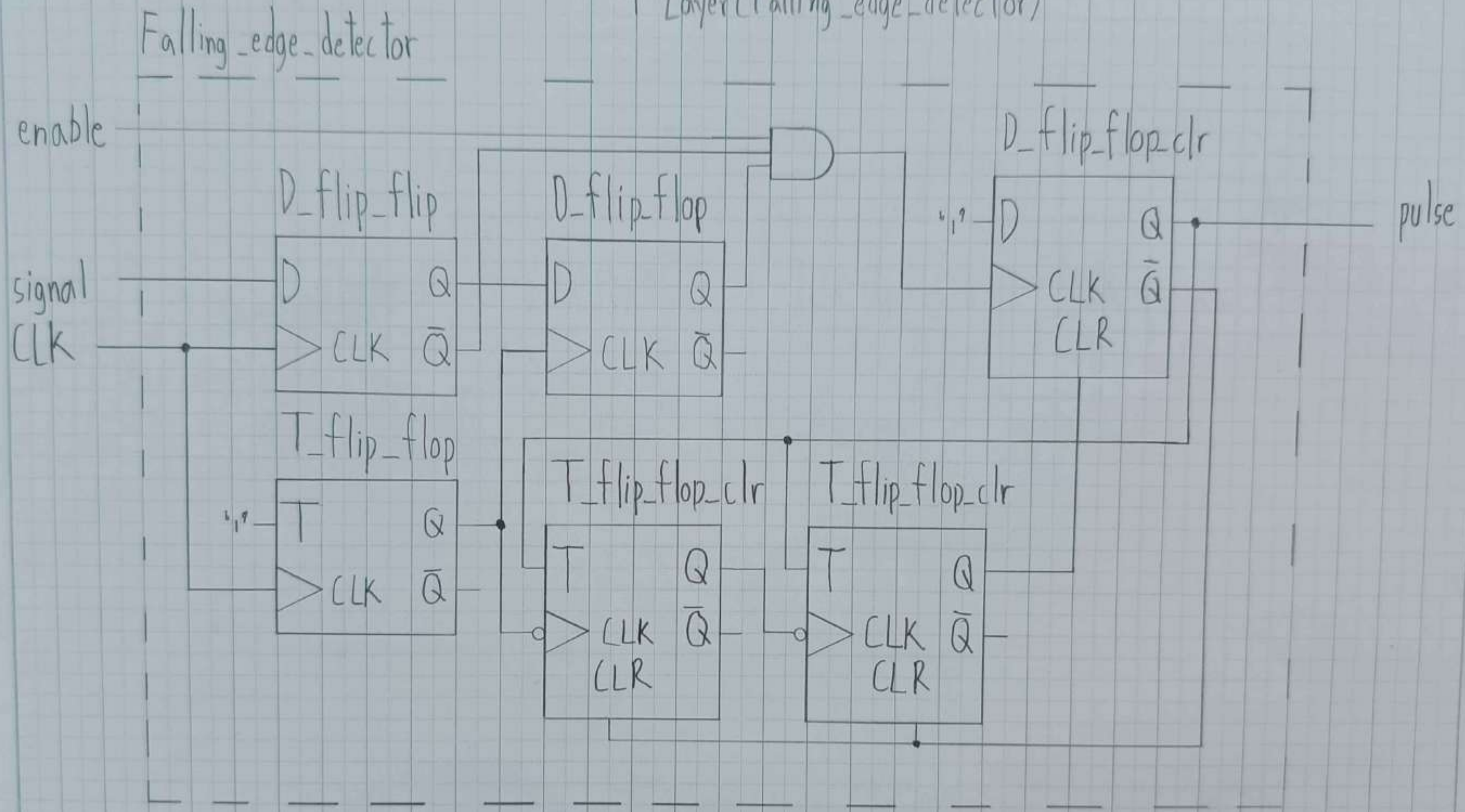
▽ Lab6 นาย อรุณ รักษ์วงศ์สิริ 68011008 sec102

4th Layer (Comparator)


Comparator
Comparator_4_bit
a[3:0]
b[3:0]
GT
EQ
"1001"
Comparator_4_bit
C_ones[3:0]
sw_ones[3:0]
a[3:0]
b[3:0]
GT
EQ
Comparator_4_bit
c_tens[3:0]
sw_tens[3:0]
a[3:0]
b[3:0]
GT
EQ
Comparator_4_bit
"1001"
a[3:0]
b[3:0]
GT
EQ
equal
error

▽ Lab6 นายวรุฒ รักษ์วงศ์สิริ 68011008 sec 102

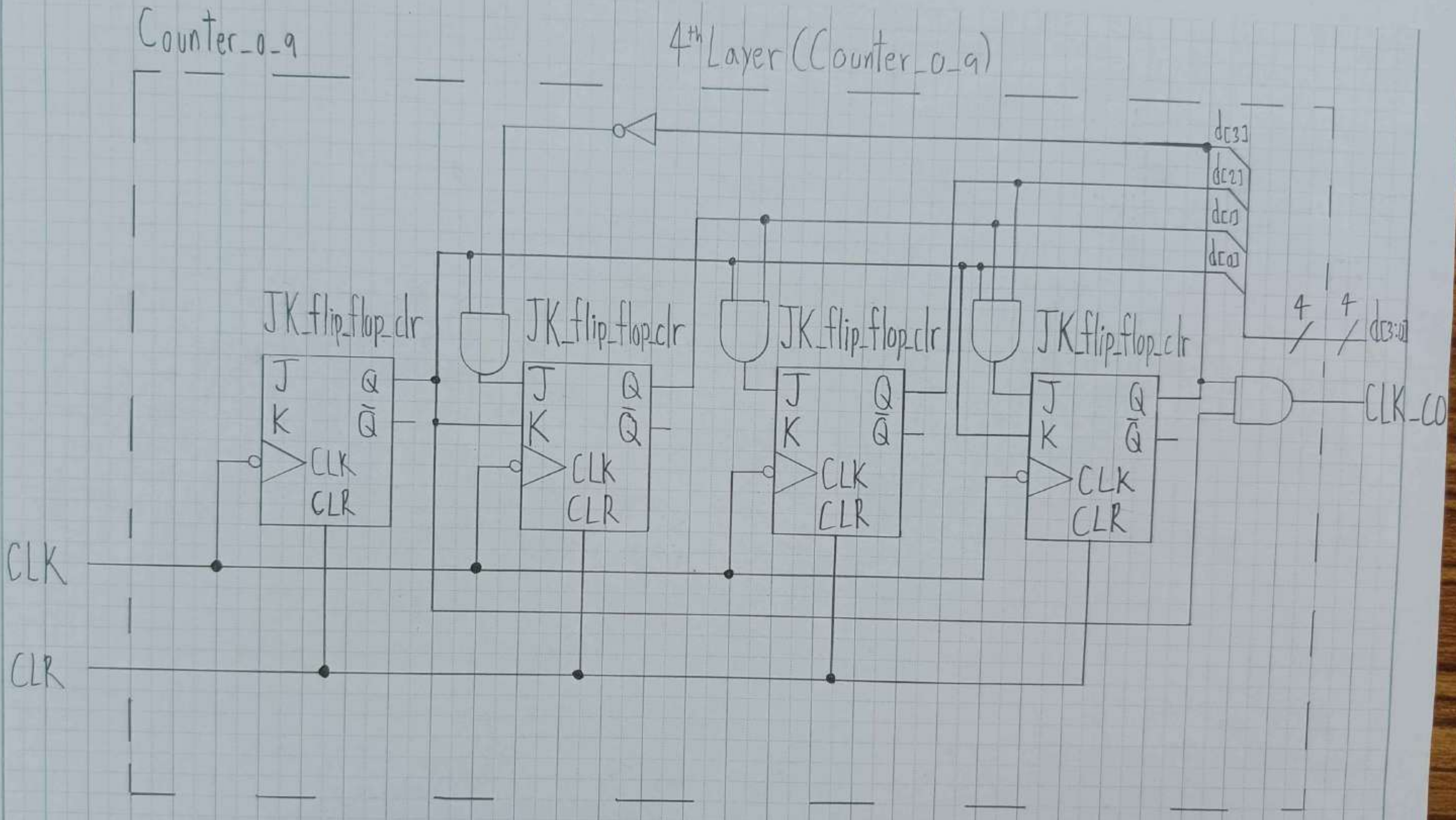
4^{th} Layer (Falling_edge_detector)


Falling_edge_detector
enable
signal
CLK
D_flip_flip
D
CLK
Q
$\bar{Q}$
D_flip_flop
D
CLK
Q
$\bar{Q}$
T_flip_flop
"1"
T
CLK
Q
$\bar{Q}$
T_flip_flop_clr
T
CLK
CLR
Q
$\bar{Q}$
T_flip_flop_clr
T
CLK
CLR
Q
$\bar{Q}$
D_flip_flop_clr
"1"
D
CLK
CLR
Q
$\bar{Q}$
pulse

Lab6 นายอรุณ รักษ์วงศ์ศิริ 6801 1008 Sec 102

NO.
DATE / /

4th Layer (Counter_0_9)


Counter_0_9
JK_flip_flop_clr
J
K
Q
$\bar{Q}$
CLK
CLR
JK_flip_flop_clr
JK_flip_flop_clr
JK_flip_flop_clr
d[3]
d[2]
d[1]
d[0]
4
4
d[3:0]
CLK_CO
CLK
CLR

Lab 6 นายอรุณ รักษ์วงศ์ศิริ 68011008 sec 10 2